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HSRP

[Hot Standby Router Protocol]

GNS3 || Cisco IOS

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Table of Contents

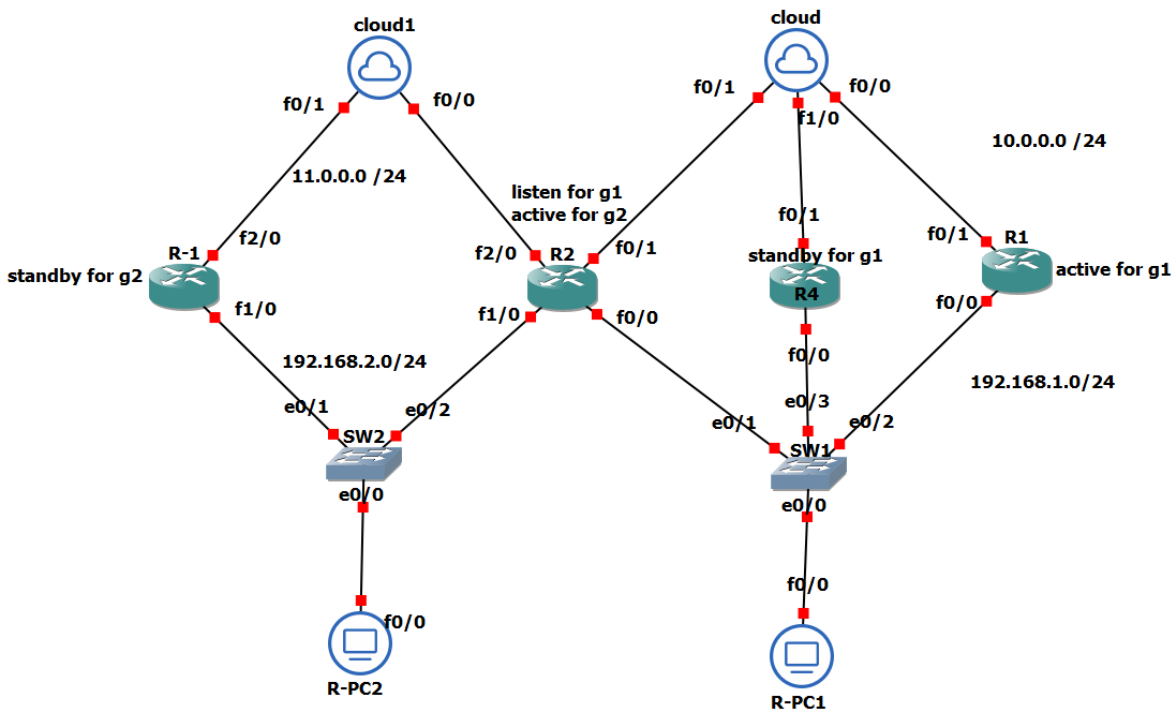
Hot Standby Router Protocol (HSRP) Lab Manual

Dual-Group Configuration, Interface Tracking & First-Hop Redundancy Analysis

1. Project Overview

This laboratory document provides a comprehensive overview and detailed technical guide for configuring Cisco's Hot Standby Router Protocol (HSRP) across a multi-router topology. The primary objective is to implement First-Hop Redundancy Protocol (FHRP) capabilities alongside Interface Tracking and Preemption to eliminate Single Points of Failure (SPOF) and achieve active/active traffic load distribution across multiple Local Area Networks (LANs).

2. Network Topology



3. Addressing Scheme

Device	Interface	IP Address / Subnet	Role / Description
R1	FastEthernet0/0 FastEthernet0/1 Loopback0	192.168.1.1 /24 10.0.0.1 /24 1.1.1.1 /32	Group 1 Active Gateway WAN 1 Link Tracked Target
R4	FastEthernet0/0 FastEthernet0/1	192.168.1.4 /24 10.0.0.4 /24	Group 1 Standby Gateway WAN 1 Link
R2	FastEthernet0/0 FastEthernet1/0 FastEthernet2/0	192.168.1.2 /24 192.168.2.2 /24 11.0.0.2 /24	Group 1 Listen Gateway Group 2 Active Gateway WAN 2 Link
R-1	FastEthernet1/0 FastEthernet2/0	192.168.2.1 /24 11.0.0.3 /24	Group 2 Standby Gateway WAN 2 Link
R-PC1	FastEthernet0/0	192.168.1.200 /24	Host in LAN 1 (Gateway: 192.168.1.3)
R-PC2	FastEthernet0/0	192.168.2.200 /24	Host in LAN 2 (Gateway: 192.168.2.3)

4. Technical Specifications & HSRP Mechanics

HSRP Group 1 Specification (LAN 1 - 192.168.1.0/24)

- Virtual IP Address: 192.168.1.3
- Primary Active Router: R1 (Configured Priority: 150)
- Primary Standby Router: R4 (Configured Priority: 101)
- Backup Listener: R2 (Default Priority: 100)
- Interface Tracking & Failover Logic: R1 tracks Loopback0 (Track 200) and FastEthernet0/1 (Track 300). If either tracked interface fails, R1 decreases its priority by 60 points. With a reduced priority of 90, R4 (Priority 101) immediately preempts and assumes the Active router role for Group 1.

HSRP Group 2 Specification (LAN 2 - 192.168.2.0/24)

- Virtual IP Address: 192.168.2.3
- Primary Active Router: R2 (Configured Priority: 105)
- Primary Standby Router: R-1 (Default Priority: 100)
- Interface Tracking & Failover Logic: R2 tracks FastEthernet2/0 (Track 500). Upon WAN link disruption on FastEthernet2/0, R2 decreases its priority by 15 points (down to 90). R-1 then preempts R2 and takes over active gateway duties for Group 2.

5. Device Configurations

Router R1 Configuration

```
hostname R1
!
track 200 interface Loopback0 line-protocol
track 300 interface FastEthernet0/1 line-protocol
!
interface Loopback0
 ip address 1.1.1.1 255.255.255.255
!
interface FastEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 standby 1 ip 192.168.1.3
 standby 1 priority 150
 standby 1 preempt
 standby 1 track 200 decrement 60
 standby 1 track 300 decrement 60
!
interface FastEthernet0/1
 ip address 10.0.0.1 255.255.255.0
!
```

Router R4 Configuration

```
hostname R4
!
track 300 interface FastEthernet0/1 line-protocol
!
interface FastEthernet0/0
 ip address 192.168.1.4 255.255.255.0
 standby 1 ip 192.168.1.3
 standby 1 priority 101
 standby 1 preempt
 standby 1 track 300 decrement 5
!
interface FastEthernet0/1
 ip address 10.0.0.4 255.255.255.0
!
```

Router R2 Configuration

```
hostname R2
!
track 500 interface FastEthernet2/0 line-protocol
!
interface FastEthernet0/0
 ip address 192.168.1.2 255.255.255.0
 standby 1 ip 192.168.1.3
 standby 1 preempt
!
interface FastEthernet1/0
 ip address 192.168.2.2 255.255.255.0
 standby 2 ip 192.168.2.3
 standby 2 priority 105
 standby 2 preempt
 standby 2 track 500 decrement 15
!
interface FastEthernet2/0
 ip address 11.0.0.2 255.255.255.0
!
```

Router R-1 Configuration

```
hostname R-1
!
interface FastEthernet1/0
 ip address 192.168.2.1 255.255.255.0
 standby 2 ip 192.168.2.3
 standby 2 preempt
!
interface FastEthernet2/0
 ip address 11.0.0.3 255.255.255.0
!
```

6. Verification and Diagnostic Commands

Use the following Cisco IOS commands to verify state transitions and interface tracking:

```
show standby brief
show standby fastEthernet 0/0
show track
```